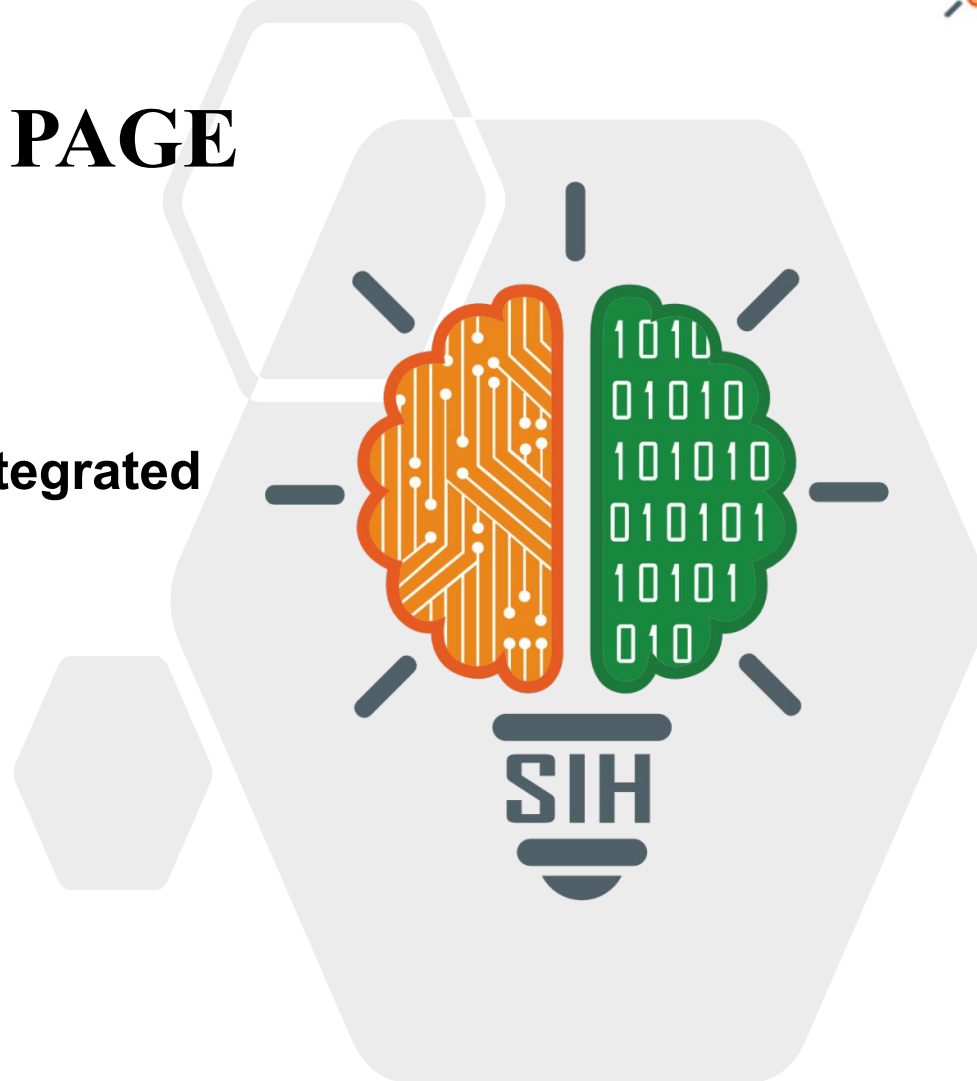


SMART INDIA HACKATHON 2025



TITLE PAGE

- Problem Statement ID – SIH25103
- Problem Statement Title- ERP-based Integrated Student Management System
- Theme- Smart Automation
- PS Category- Software
- Team ID-
- Team Name (Registered on portal)



IDEA / SOLUTION :

Implementation of a **Comprehensive ERP suite** that can **unify data** and **remain in financial reach** for all public colleges:

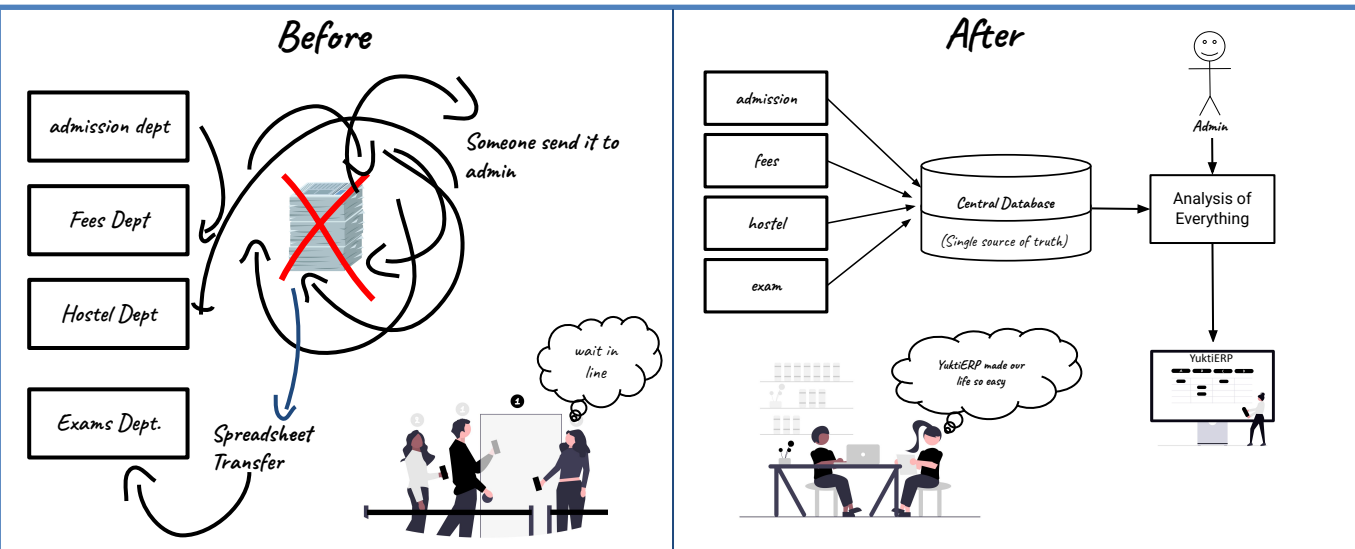
- ❖ **Integrated Workflow** in which data flows seamlessly into a **central database**.
- ❖ **UI based on familiar tools** ensures adoption.
- ❖ **Role-based access control** ensures institutional security.
- ❖ Built-in **privacy** with complete **data ownership**.
- ❖ Capability to work in **Real-time** enabling **transparent administration**.

Problem Resolution :

- ❖ **Low-cost ERP** integrates admissions, fees, hostel, and exam data, creating a **single source of truth without heavy infrastructure**.
- ❖ **Eliminates** manual data entry, redundant records, and fragmented ledgers **by providing real-time insights and secure, unified workflows**.

Unique Value Propositions (UVP) :

- ❖ **Dashboard for every user** – personalized insights for students, staff, and admins
- ❖ **Instant data retrieval** – index-based search fetches results in under a second
- ❖ **Hybrid database with sharding** – structured + unstructured data stored efficiently
- ❖ **API-driven** modularization for flexible college ERP



TECH STACK

Algorithm Development:

Next.js & Express.js - Core technologies used for developing the **YuktiERP**.

Mobile Application Development:

Flutter - Framework used for building the mobile application, ensuring **cross-platform compatibility**.

Encryption and Security:

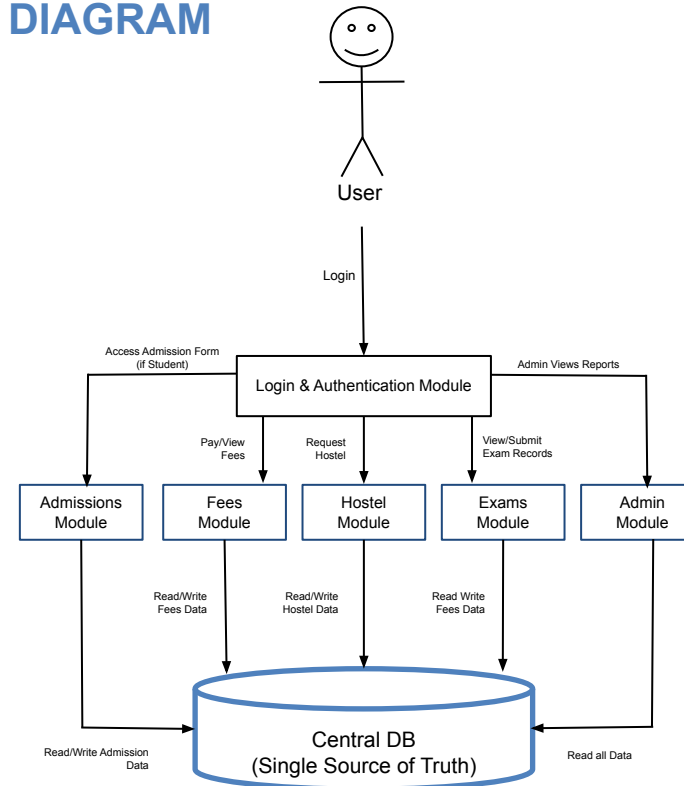
Secure data with **UUID keys**, enforced relations, **strict input validation**, and **Role Based Access**

Cloud Services:

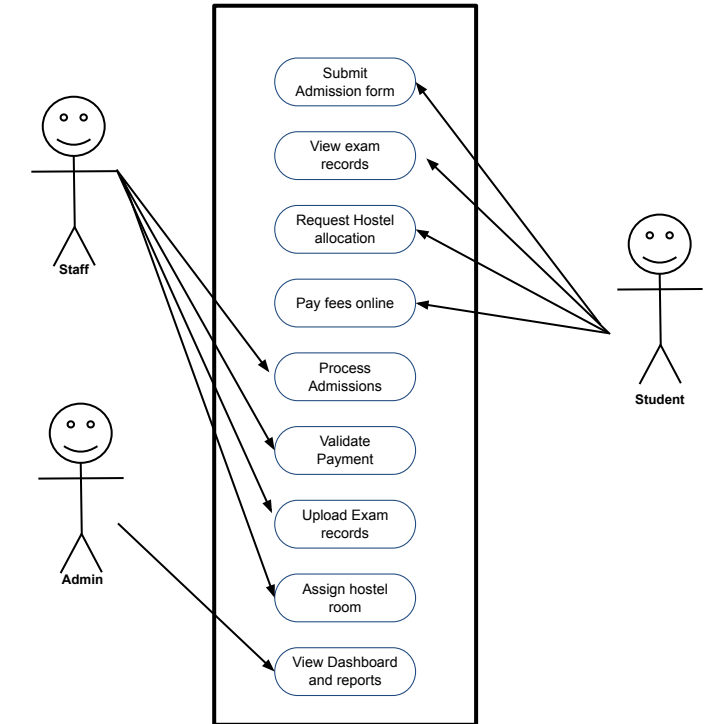
Hybrid DB Structure: MongoDB + PostgreSQL

Docker: For easy deployment.

SYSTEM ARCHITECTURE DIAGRAM



USE CASE DIAGRAM



Product Status: 80% product built completed and further build is on progress. Testing and validation process are next to be undergone

FEASIBILITY AND VIABILITY :

- ❖ **Technical:** Optimized offline-first system handles **10,000+ users** with **<200 ms** response time.
- ❖ **Cost reduced by ~78%:** From ₹8L/year to just ₹1.75L/year.
- ❖ **Market: 27.7% EdTech growth, 200+ colleges** ready for adoption (starting with Tier-2/3).
- ❖ **Operational:** Rollout in **2–6 weeks**, saves **75% admin time**, achieves **90% staff adoption**.

POTENTIAL CHALLENGES AND RISKS :

- ❖ **Offline-first** setup may **fail** in low-network areas.
- ❖ ₹31K/year **pricing** may **limit sustainable growth**.
- ❖ **Tier-2/3** colleges may **resist digital change** due to legacy systems.
- ❖ **Expanding** to 100+ colleges strains team capacity, **slowing deployment**.

STRATEGIES FOR OVERCOMING THESE CHALLENGES:

- ❖ **Background sync and caching** ensure reliable offline operations.
- ❖ **Tiered pricing** with add-ons sustains **long-term growth**.
- ❖ Tutorials, **multilingual guides**, and trainers boost adoption speed.
- ❖ **Automated support** and onboarding streamline large-scale rollouts.



Phase 1:
Awareness
and Outreach

Phase 2:
Onboarding &
Training

Phase 3:
Phased Rollout
(2-6 Weeks)

Phase 4:
Rollout
(1 Weeks)

Full Adoption
and Scaled
Support

IMPACT

- ❖ **Students:**
No more standing in **long queues** everything is processed faster with **less hassle**.
- ❖ **Faculty/Staff:**
No need to **re-enter** the same **data** tasks are quicker and **error-free**.
- ❖ **Administrators:**
A clear, **real-time dashboard** of the entire institution helps in making **smarter decisions**.

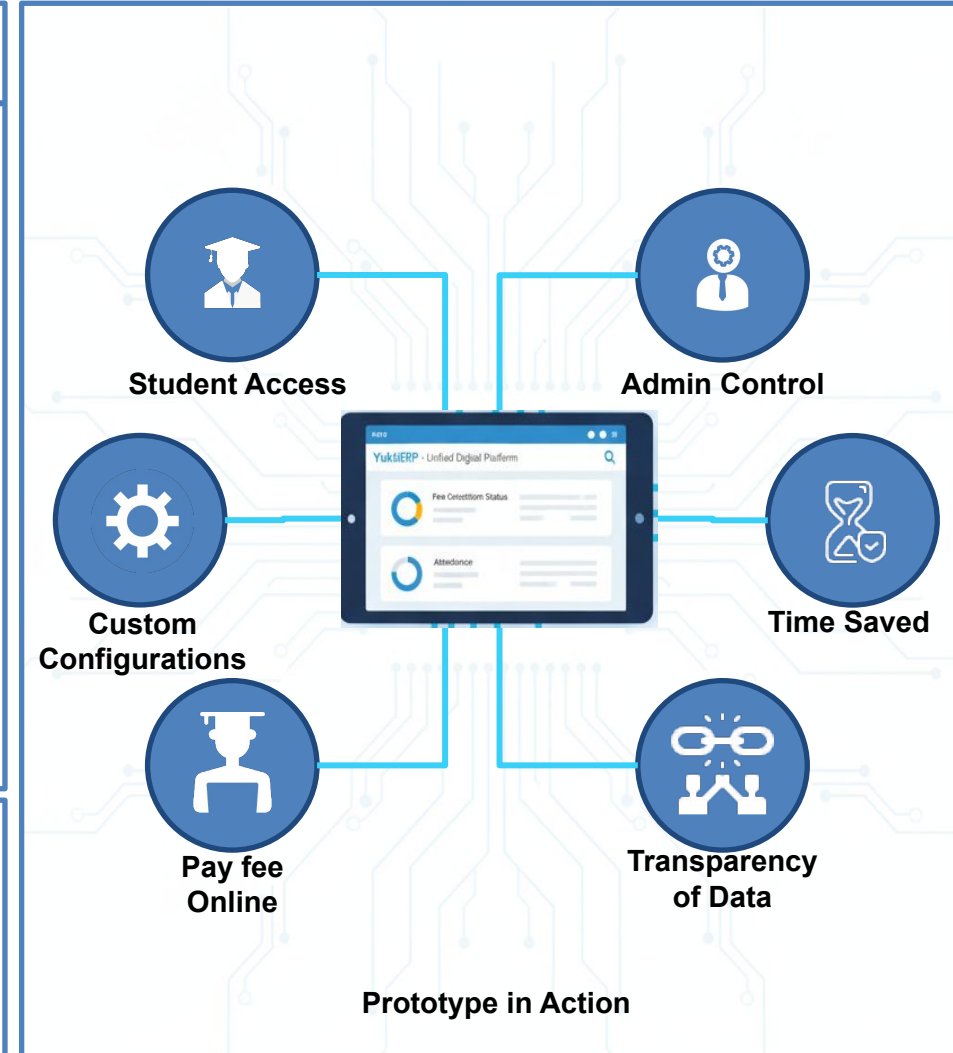
BENEFITS

- ❖ **Students:**
Save valuable time and enjoy seamless convenience for studies, extracurriculars, and campus life.
- ❖ **Faculty/Staff:**
Focus energy on teaching, mentoring, and innovation instead of administrative work.
- ❖ **Administrators:**
Ensure efficient resource allocation, hassle-free compliance, and a stronger institutional reputation.

Unified Platform: Our solution acts as a single source of truth, ending fragmented operations.

Seamless Data Flow: Enables real-time data flow from a student's admission to their graduation.

Empowered Institution: Empowers the entire institution with unparalleled efficiency.



Impacts of Digital Technologies on Education and Factors Influencing Adoption –

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9684747/>

Reference: Enhancing Student Data Management with a Smart ERP System –

<https://thecodework.com/blog/enhancing-student-data-management-with-a-smart-erp-system/>

Reference: How Universities Can Save 20% Operational Costs with ERP Software –

<https://www.academiaerp.com/blog/how-universities-can-save-20-operational-costs-with-erp-software/>

Reference: India Education ERP Market Size & Outlook, 2024–2030 –

<https://www.grandviewresearch.com/horizon/outlook/education-erp-market/india>

Reference: Re-envisioning Higher Education Through Digital Adoption –

<https://www.pwc.in/assets/pdfs/industries/re-envisioning-higher-education-through-digital-adoption.pdf>

Reference: Digital Transformation in Higher Education in 2025 – <https://whatfix.com/blog/digital-transformation-in-higher-education/>

Reference: Education ERP Global Market Report 2025 –

<https://www.giiresearch.com/report/tbrc1659019-education-erp-global-market-report.html>

Reference: 12 Reasons Why Your School Should Go Paperless –

<https://www.tads.com/how-tads-paperless-system-helps-the-environment/>